Battery switch, when placed in the BAT ON/EPU OFF position, connects the onboard battery to the Starter and Main Buses. Battery charging is then supplied by the generators when they are brought on-line. When the battery switch is placed in the BAT OFF/EPU ON position a 28V DC external power unit (EPU) may then supply power to the Starter Bus which in turn powers both the Main and Auxiliary Buses. An EPU ON caution light comes on when external power is connected and switched on. The EPU receptacle in the nose compartment allows for a starting current of 700 A.

BAT 60 and BAT 70 warning lights are provided to indicate an overheating condition of the battery. These lights operate independently being activated by separate battery-mounted temperature sensors.

Voltmeter (DC VOLTS) indicator indicates the voltage output supplied by the Main Bus during generator, battery or external power operation.

Ammeter (DC AMPS) indicator with the CURRENT IND switch placed in the BUS BAR position indicates current consumption (– side) measured between the Starter Bus and Main Bus, and with the CURRENT IND switch in either 1 GEN or 2 GEN position will indicate the current supplied (+ side) by the respective generator.

7.11.4 Circuit Protection

All electrical power circuits are protected by either a circuit breaker or fuse. Most circuit breakers are located on the center console switch panel. Most low amperage fuses are found on a fuse panel (see Fig. 7-18) located behind a cover on the instrument panel lower RH structure. Fuses and circuit breakers used to protect high amperage consumers and the generators are found in a main relay box mounted to the fuselage RH structure in the cargo compartment.

7.12 LIGHTING EQUIPMENT

The lighting equipment comprises exterior and interior lighting systems and the annunciator lights. The exterior lighting system includes the position lights, the landing light and the anticollision lights. The interior lighting system includes the instrument lights, a utility light, and the internal compartment lights.

7.12.1 Position Lights

The position lights are located on the horizontal stabilizer end-plates (RH - green, LH - red) and on the top aft end of the vertical fin (white). Power for operation of the lights is provided by the Main Bus via a fuse. Operation of the lights is controlled by the POSITION LIGHT switch on the overhead console.

7.12.2 Anticollision Lights

Two high-intensity flashing red anticollision lights are located, one on top of the tail rotor gearbox, the other on the underside of the fuselage. Power for operation of the lights is provided by the Main Bus via a fuse. Operation of the lights is controlled by the ANTI-COLL LIGHT switch on the overhead console.

7.12.3 **Instrument Lights**

Power to the instrument lights is controlled by the INSTR LIGHT switch on the center console switch panel. With this switch on power is supplied from the Main Bus via a fuse to a dim control unit. The light intensity is then controlled by the INSTR LIGHT rheostat switch located on the instrument panel.

A whip light at the left of the center console switch panel is also powered when the INSTR LIGHT switch is on. The whip light, however, has its own on-off switch and the light intensity is not effected by the rheostat switch.

7.12.4 **Utility Light**

A utility light is provided as a detachable light with coiled extension cord located at the left of the overhead console. This light is operated by an OFF-BRT rheostat switch integral with the light unit and is independently powered by the Main Bus via a fuse.

7.12.5 **Interior Lights**

A cabin light and two cargo compartment lights are powered from the Main Bus via the same fuse provided for the utility light. Operation of the three ceiling lights is controlled by the INTERNAL LIGHT switch on the overhead console.

7.12.6 **Annunciator Lights (see Fig. 7-19)**

When the TEST WARN. LIGHTS pushbutton on the instrument panel is pressed, all warning/caution lights (except the two F [Fire] and the LIMIT [MM ind] lights) will come on. Each warning/caution light segment has two lamps so that failure of a single lamp will produce only a dimming effect. If a segment fails to light, it indicates either the failure of both lamps or a defective circuit.

7.12.7 **Rpm Warning System**

Visual and in most cases audio signals are provided as a warning whenever:

– N_{Ro} is above or below the normal operating range,

– an engine N_1 differential equal to or greater than 12 % occurs (after either power lever is in the FLIGHT position).

The engine N_1 differential warning may be cancelled by pressing the RPM WARN pushbutton on the cyclic stick grip.

WORD SEGMENT	COLOR	ILLUMINATION PARAMETER OR FAULT
ANTI ICING 1	Amber	No. 1 engine anti-icing actuator in open position.
ANTI ICING 2	Amber	No. 2 engine anti-icing actuator in open position.
BAT 60	Red	Battery temperature exceeds 60 °C.
BAT 70	Red	Battery temperature exceeds 70 °C.
EPU ON	Amber	External power unit connected and switched on.
F (LH)	Red	Fire detected in the No. 1 engine compartment.
F (RH)	Red	Fire detected in the No. 2 engine compartment.
FILT 1	Red	No. 1 fuel filter is clogged and bypass has begun.
FILT 2	Red	No. 2 fuel filter is clogged and bypass has begun.
GEN 1	Red	No. 1 generator is not on-line.
GEN 2	Red	No. 2 generator is not on-line.
HY BLOCK	Red	Jammed servo valve in hydraulic system no. 1 and cross-over to system no. 2 has occurred.
HYD 1	Red	No. 1 hydraulic system pressure is low and cross-over to system no. 2 has occurred.
HYD 2	Red	No. 2 hydraulic system pressure is low.
LIMIT	Red	Max mast moment limit exceeded.
LOW FUEL	Red	Fuel supply tank usable quantity is below 60 kg.
MAG PLUG 1	Red	No. 1 engine oil contains metal fragments.
MAG PLUG 2	Red	No. 2 engine oil contains metal fragments.
OIL COOL	Red	(After SB 60-37) Oil cooler fan failure.
RPM	Red	N_{Ro} below 95% or above 103%; N_1 differential greater than 12%.
T OIL	Red	Transmission oil pressure below 0.5 kp/cm^2; or oil temperature above 105 °C.

Fig. 7-19 Alphabetical Listing of Warning/Caution Lights